

Proof Appendix: Sub-Gaussian and Central-Condition Inequalities

1 Setup and notation

Let $\mathcal{Z} := \mathcal{X} \times \mathcal{Y}$ and $\mathcal{D} := \mathcal{Z}^n$ be the dataset space. Let $D_n \in \mathcal{D}$ denote the true dataset and $\hat{D}_n \in \mathcal{D}$ denote the pseudo dataset. Let f_T be the teacher model and f_S be the student model. Define the population and empirical risks

$$L_P(f) := \mathbb{E}_{(X,Y) \sim P}[\ell(f; X, Y)], \quad L_{D_n}(f) := \frac{1}{n} \sum_{i=1}^n \ell(f; z_i), \quad L_{\hat{D}_n}(f) := \frac{1}{n} \sum_{i=1}^n \ell(f; \hat{z}_i).$$

Define the generalization errors

$$\text{gen}_T := \mathbb{E}_{D_n, f_T} [L_P(f_T) - L_{D_n}(f_T)], \quad \text{gen}_S := \mathbb{E}_{\hat{D}_n, f_S} [L_P(f_S) - L_{\hat{D}_n}(f_S)].$$

Let Δ_{D_n, f_T} and $\Delta_{\hat{D}_n, f_S}$ be the corresponding process-level joint laws. Assume $\Delta_{\hat{D}_n, f_S} \ll \Delta_{D_n, f_T}$ and define the process-level KL

$$\mathsf{K}_n := \text{KL}(\Delta_{\hat{D}_n, f_S} \| \Delta_{D_n, f_T}).$$

Finally define

$$h(d, f) := L_P(f) - L_d(f), \quad H := h(D_n, f_T) = L_P(f_T) - L_{D_n}(f_T).$$

Then $\mathbb{E}_{\Delta_{D_n, f_T}}[H] = \text{gen}_T$ and $\mathbb{E}_{\Delta_{\hat{D}_n, f_S}}[h] = \text{gen}_S$.

2 Donsker–Varadhan bound

By the Donsker–Varadhan variational inequality, for any measurable g ,

$$\mathbb{E}_{\Delta_{\hat{D}_n, f_S}}[g] - \log \mathbb{E}_{\Delta_{D_n, f_T}}[e^g] \leq \text{KL}(\Delta_{\hat{D}_n, f_S} \| \Delta_{D_n, f_T}) = \mathsf{K}_n.$$

Taking $g = \lambda h$ yields, for any $\lambda > 0$,

$$\lambda \text{gen}_S - \log \mathbb{E}_{\Delta_{D_n, f_T}}[e^{\lambda h(D_n, f_T)}] \leq \mathsf{K}_n, \tag{1}$$

hence

$$\text{gen}_S \leq \frac{\mathsf{K}_n}{\lambda} + \frac{1}{\lambda} \log \mathbb{E}_{\Delta_{D_n, f_T}}[e^{\lambda H}]. \tag{2}$$

3 Sub-Gaussian upper inequality

Theorem 1 (Sub-Gaussian inequality). *Assume H is σ^2 -sub-Gaussian under Δ_{D_n, f_T} , i.e.,*

$$\log \mathbb{E}_{\Delta_{D_n, f_T}} [e^{\lambda(H - \mathbb{E}H)}] \leq \frac{\lambda^2 \sigma^2}{2}, \quad \forall \lambda \in \mathbb{R}.$$

Then

$$\text{gen}_S - \text{gen}_T \leq \sigma \sqrt{2K_n}.$$

证明. By sub-Gaussianity,

$$\log \mathbb{E}[e^{\lambda H}] = \log \mathbb{E}[e^{\lambda(H - \mathbb{E}H)}] + \lambda \mathbb{E}H \leq \frac{\lambda^2 \sigma^2}{2} + \lambda \text{gen}_T.$$

Plugging into (2) gives, for any $\lambda > 0$,

$$\text{gen}_S \leq \text{gen}_T + \frac{K_n}{\lambda} + \frac{\lambda \sigma^2}{2}.$$

Optimize over $\lambda > 0$: the minimum is achieved at $\lambda^* = \sqrt{2K_n}/\sigma$. Substituting yields

$$\text{gen}_S - \text{gen}_T \leq \sigma \sqrt{2K_n}.$$

□

3.1 A concrete σ^2 from bounded loss and stability (optional)

Proposition 1 (Stability implies sub-Gaussianity). *Assume the loss is bounded: $\ell(f; z) \in [a, b]$ for all f, z . Let the teacher training algorithm $A_T : \mathcal{Z}^n \rightarrow \mathcal{F}_T$ be β -uniformly stable: for any neighboring datasets $D, D^{(i)}$,*

$$\sup_{z \in \mathcal{Z}} |\ell(A_T(D); z) - \ell(A_T(D^{(i)}); z)| \leq \beta.$$

Define $H(D) := L_P(A_T(D)) - L_D(A_T(D))$. Then $H(D) - \mathbb{E}[H(D)]$ is sub-Gaussian with variance proxy

$$\sigma^2 = \frac{n}{4} \left(2\beta + \frac{b-a}{n} \right)^2.$$

证明. For neighboring $D, D^{(i)}$,

$$\begin{aligned} |H(D) - H(D^{(i)})| &\leq |L_P(A_T(D)) - L_P(A_T(D^{(i)}))| + |L_D(A_T(D)) - L_{D^{(i)}}(A_T(D^{(i)}))| \\ &\leq \beta + \left(|L_D(A_T(D)) - L_D(A_T(D^{(i)}))| + |L_D(A_T(D^{(i)})) - L_{D^{(i)}}(A_T(D^{(i)}))| \right) \\ &\leq \beta + \beta + \frac{b-a}{n} = 2\beta + \frac{b-a}{n}. \end{aligned}$$

Thus the bounded difference constant is $c_i = 2\beta + \frac{b-a}{n}$ for all i . McDiarmid's mgf form yields sub-Gaussianity with

$$\sigma^2 = \frac{1}{4} \sum_{i=1}^n c_i^2 = \frac{n}{4} \left(2\beta + \frac{b-a}{n} \right)^2.$$

□

4 Central condition lower inequality

Definition 1 ((η, c)-central condition). Let $\eta > 0$ and $0 < c \leq 1$. A random variable X with law P satisfies the (η, c)-central condition if

$$\log \mathbb{E}_P[e^{-\eta X}] \leq -c\eta \mathbb{E}_P[X],$$

given $0 < \mathbb{E}_P[X] < \infty$.

Theorem 2 (Central-condition inequality). Assume H satisfies the (η, c)-central condition under Δ_{D_n, f_T} :

$$\log \mathbb{E}_{\Delta_{D_n, f_T}}[e^{-\eta H}] \leq -c\eta \text{gen}_T.$$

Then

$$\text{gen}_S \geq c \text{gen}_T - \frac{1}{\eta} \mathsf{K}_n.$$

Equivalently,

$$\text{gen}_T \leq \frac{1}{c} \text{gen}_S + \frac{1}{c\eta} \mathsf{K}_n.$$

证明. Apply Donsker–Varadhan with $g = -\eta h$:

$$\mathbb{E}_{\Delta_{\hat{D}_n, f_S}}[-\eta h] - \log \mathbb{E}_{\Delta_{D_n, f_T}}[e^{-\eta h}] \leq \mathsf{K}_n.$$

Thus

$$-\eta \text{gen}_S \leq \mathsf{K}_n + \log \mathbb{E}_{\Delta_{D_n, f_T}}[e^{-\eta H}] \leq \mathsf{K}_n - c\eta \text{gen}_T,$$

so

$$\text{gen}_S \geq c \text{gen}_T - \frac{1}{\eta} \mathsf{K}_n.$$

The equivalent form follows by rearranging. □

4.1 Sub-Gaussianity implies a central condition (optional)

Lemma 1 (Parameter relation). Let H be σ^2 -sub-Gaussian with mean $\mu = \mathbb{E}[H] > 0$. Then for any $\eta > 0$,

$$\log \mathbb{E}[e^{-\eta H}] \leq \frac{\eta^2 \sigma^2}{2} - \eta \mu.$$

Hence the (η, c)-central condition holds whenever

$$c \leq 1 - \frac{\eta \sigma^2}{2\mu}, \quad \eta < \frac{2\mu}{\sigma^2}.$$

证明.

$$\log \mathbb{E}[e^{-\eta H}] = \log \mathbb{E}[e^{-\eta(H-\mu)}] - \eta \mu \leq \frac{\eta^2 \sigma^2}{2} - \eta \mu.$$

The stated condition follows by requiring $\frac{\eta^2 \sigma^2}{2} - \eta \mu \leq -c\eta \mu$. □